

S.No	Particulars	Theory		Practical	
		Amplitude	Frequency	Amplitude	Time Frequency
1.	Message Signal	3 V	300 Hz	3 V	$T_m = x_2 - x_1$ $= 0.00335$ 300 Hz
2.	Carrier Signal	5 V	3000 Hz	5 V	$T_m = x_2 - x_1$ $= 0.000333$ 3000 Hz
3.	AM Wave	-		$E_{max} = 8 V$ $E_{min} = 2 V$	$T_{AM} = x_2 - x_1$ $= 0.00033$ 3000 Hz

Ex: NO: 1
21/07/26

SIMULATION OF AMPLITUDE MODULATION (AM) AND DEMODULATION OF MODULATION INDEX AND BANDWIDTH ANALYSIS

AIM:

To generate and analysis an amplitude modulated (AM) wave using SCILAB and observe the message signal, carrier signal and wave form.

APPARATUS REQUIRED:

- * Laptop with i3 processor
- * SCILAB software.

CODING:

clc;

clear all;

$E_m = 10;$

$F_m = 300;$

$E_c = 50;$

$F_c = 3000;$

$F_s = 30000;$

$t = 0: 2/F_s: 3/F_m;$

$am = E_m * \sin(2 * \% 3.14 * F_m * t)$

Calculation:

1. Message Signal:

Practical time

$$T_m = x_2 - x_1$$

$$x_2 = 0.00333$$

$$x_1 = 0.5$$

$$T_m = 0.00333$$

Practical frequency:

$$f_m = 1/T_m$$

$$= 1/0.00333$$

$$= 300 \text{ Hz}$$

2. Carrier Signal:

Practical time

$$T_c = x_2 - x_1$$

$$x_2 = 0.000333$$

$$x_1 = 0.05$$

$$T_c = 0.000333$$

Practical frequency:

$$f_c = 1/T_c$$

$$f_c = 1/0.000333$$

$$f_c = 3000 \text{ Hz}$$

```
subplot (3, 1, 1);
```

```
plot (t, am);
```

```
title ("Message signal");
```

```
c = Ec * sin (2 * pi * 3.14 * Fc * t);
```

```
subplot (3, 1, 2);
```

```
plot (t, c);
```

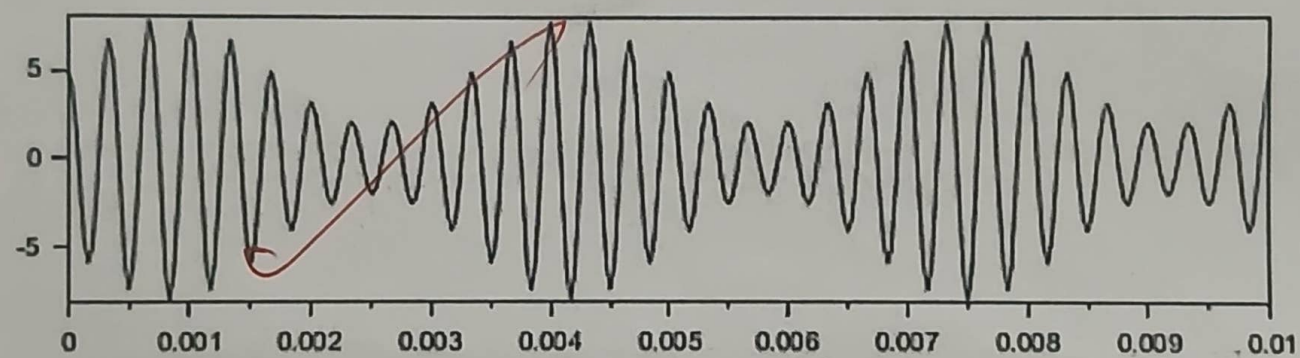
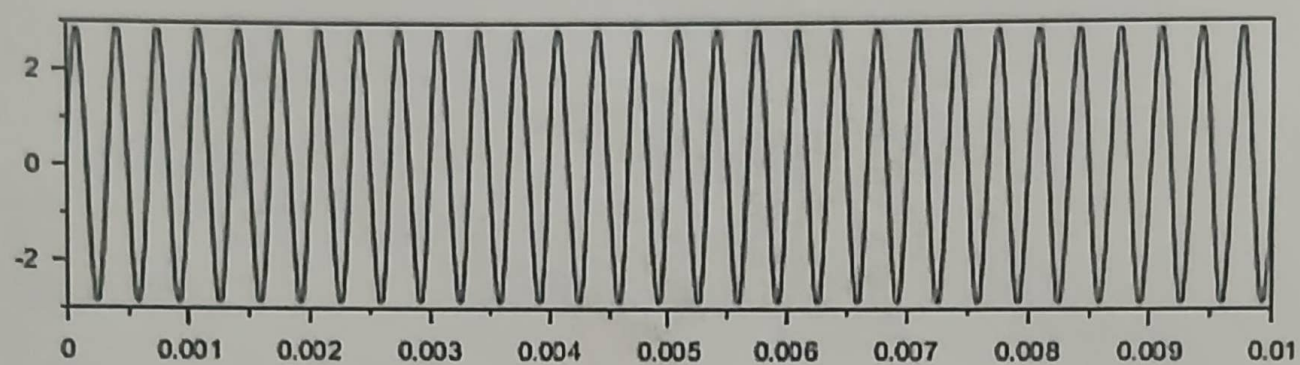
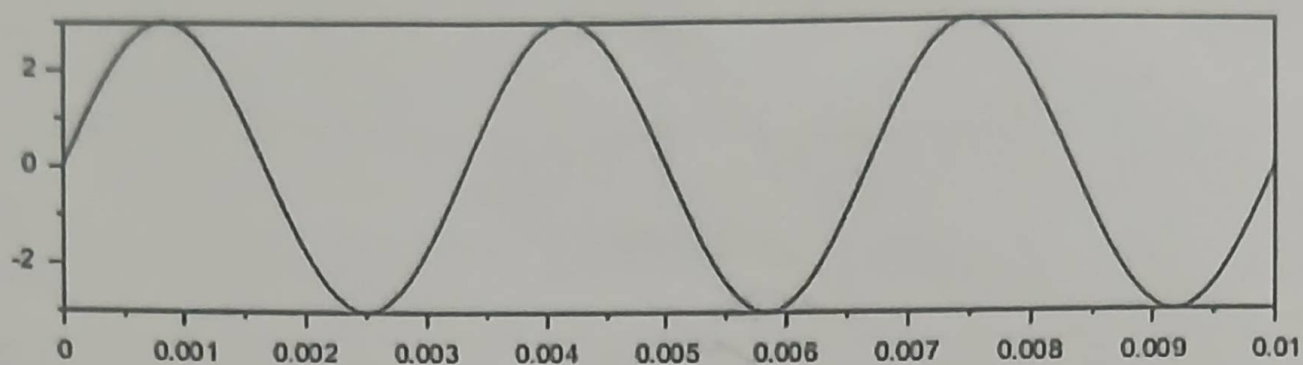
```
title ("Carrier signal");
```

```
e am = (Ec + am) .* cos (2 * pi * 3.14 * Fc * t);
```

```
subplot (3, 1, 3);
```

```
plot (t, eam);
```

```
title ("Am Wave");
```



Scilab Program Implement (20)	Wave form & tabulation (20)	Calculation & Result (20)	Viva voice (20)	Record observation (20)	Total mark
20	20	20	16	20	96

RESULT :

Thus the Amplitude Modulated (AM) ~~Signal~~ was Successfully done using SCI LAB